



May 9, 2025

Pearl Inc.
% Ashley Brown
Official Correspondent
2515 Benedict Canyon Dr.
BEVERLY HILLS, CA 90210

Re: K243230
Trade/Device Name: Second Opinion® BLE
Regulation Number: 21 CFR 892.2050
Regulation Name: Medical Image Management And Processing System
Regulatory Class: Class II
Product Code: QIH
Dated: April 3, 2025
Received: April 3, 2025

Dear Ashley Brown:

We have reviewed your section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (the Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database available at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Additional information about changes that may require a new premarket notification are provided in the FDA guidance documents entitled "Deciding When to Submit a 510(k) for a Change to an Existing Device" (<https://www.fda.gov/media/99812/download>) and "Deciding When to Submit a 510(k) for a Software Change to an Existing Device" (<https://www.fda.gov/media/99785/download>).

Your device is also subject to, among other requirements, the Quality System (QS) regulation (21 CFR Part 820), which includes, but is not limited to, 21 CFR 820.30, Design controls; 21 CFR 820.90, Nonconforming product; and 21 CFR 820.100, Corrective and preventive action. Please note that regardless of whether a change requires premarket review, the QS regulation requires device manufacturers to review and approve changes to device design and production (21 CFR 820.30 and 21 CFR 820.70) and document changes and approvals in the device master record (21 CFR 820.181).

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR Part 803) for devices or postmarketing safety reporting (21 CFR Part 4, Subpart B) for combination products (see <https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR Part 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR Parts 1000-1050.

All medical devices, including Class I and unclassified devices and combination product device constituent parts are required to be in compliance with the final Unique Device Identification System rule ("UDI Rule"). The UDI Rule requires, among other things, that a device bear a unique device identifier (UDI) on its label and package (21 CFR 801.20(a)) unless an exception or alternative applies (21 CFR 801.20(b)) and that the dates on the device label be formatted in accordance with 21 CFR 801.18. The UDI Rule (21 CFR 830.300(a) and 830.320(b)) also requires that certain information be submitted to the Global Unique Device Identification Database (GUDID) (21 CFR Part 830 Subpart E). For additional information on these requirements, please see the UDI System webpage at <https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/unique-device-identification-system-udi-system>.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance>) and CDRH Learn (<https://www.fda.gov/training-and-continuing-education/cdrh-learn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory->

[assistance/contact-us-division-industry-and-consumer-education-dice](#)) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

A handwritten signature in black ink that reads "Lu Jiang" is positioned over a large, light blue, semi-transparent watermark of the letters "FDA".

Lu Jiang, Ph.D.
Assistant Director
Diagnostic X-Ray Systems Team
DHT8B: Division of Radiological Imaging
Devices and Electronic Products
OHT8: Office of Radiological Health
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

Enclosure

Indications for Use

510(k) Number (if known)

K243230

Device Name

Second Opinion® BLE

Indications for Use (Describe)

Second Opinion® BLE is a radiological automated image processing software device intended to identify and display bone level measurements in bitewing and periapical radiographs. It should not be used in lieu of full patient evaluation or solely relied upon to make or confirm a diagnosis.

It is designed to aid dental health professionals to review bitewing and periapical radiographs of permanent teeth in patients 12 years of age or older as a concurrent and second reader.

Type of Use (Select one or both, as applicable)

☒ Prescription Use (Part 21 CFR 801 Subpart D)

☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

This section applies only to requirements of the Paperwork Reduction Act of 1995.

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510(K) SUMMARY

1. Submitter's Identification

Pearl Inc.
2515 Benedict Canyon Dr.
Beverly Hills, CA, 90210
USA
(239) 450-8829

Contact Person: Ashley Brown Position: Director of Regulatory Affairs
Date Summary Prepared: May 9th, 2025

2. Trade Name of the Device

Second Opinion® BLE

3. Common or Usual Name

Radiological Automated Image Processing System

4. Classification Name, Regulatory Classification & Product Code

Classification Name: Medical Image Management and Processing System
Regulatory Classification: 21CFR 892.2050, Class II
Product Code: QIH (Radiological Automated Image Processing System)

5. Predicate Information

Predicate device: The primary proposed **predicate** device for clearance of Second Opinion® BLE is Overjet Dental Assist by Overjet Inc., cleared in (K210187, May 19, 2021), a semi-automated software device classified as a Class II device pursuant to 21 CFR §892.2050 Medical image management and processing system under product code LLZ.

Overjet Dental Assist is a radiological semi-automated image processing software device intended to aid dental professionals in the measurements of mesial and distal bone levels associated with each tooth from bitewing and periapical radiographs.

The Overjet Dental Assist device has the same technology (AI software) and intended use as automated image processing to aid dental professionals in the measurement and visualization of bone levels associated with bitewing and periapical radiographs.

A reference device for the clearance of Second Opinion® BLE is Videa Perio Assist, a radiological semi-automated image processing software device classified as a



Class II device pursuant to 21 CFR §892.2050 Medical image management and processing system under product code QIH. The reference device was cleared under K223296 on February 6 2023. The reference device is to justify the device's safety and effectiveness by providing benchmarks for performance testing.

The device is a radiological semi-automated image processing software device intended to aid dental professionals in the measurements and visualization of mesial and distal bone levels associated with each tooth from bitewing and periapical radiographs. Measurements are made available as linear distances or relative percentages.

The subject device is substantially equivalent to the predicate as well as the reference device as the overall intended use and nature of the software remains the same. The difference with the subject device is the addition of the bone level measurement feature.

6. Device Description

Second Opinion BLE is a radiological automated image processing software device intended to identify and display bone level measurements in bitewing and periapical radiographs. It should not be used in lieu of full patient evaluation or solely relied upon to make or confirm a diagnosis.

It is designed to aid dental health professionals to review bitewing and periapical radiographs of permanent teeth in patients 12 years of age or older as a concurrent and second reader.

Second Opinion BLE consists of three parts:

- Application Programming Interface ("API")
- Machine Learning Modules ("ML Modules")
- Client User Interface (UI) ("Client")

The processing sequence for an image is as follows:

1. Images are sent for processing via the API
2. The API routes images to the ML modules
3. The ML modules produce detection output
4. The UI renders the detection output

The API serves as a conduit for passing imagery and metadata between the user interface and the machine learning modules. The API sends imagery to the machine learning modules for processing and subsequently receives metadata generated by the machine learning modules which is passed to the interface for rendering.

Second Opinion BLE uses machine learning to detect bone level measurements. Images received by the ML modules are processed yielding detections which are represented as metadata. The final output is made accessible to the API for the purpose of sending to the UI for visualization. Detected bone level measurements are displayed as linear overlays



atop the original radiograph which indicate to the practitioner which regions contain which detected potential conditions that may require clinical review. The clinician can toggle over the image to highlight a potential condition for viewing.

7. Indications for Use

Second Opinion® BLE is a radiological automated image processing software device intended to identify and display bone level measurements in bitewing and periapical radiographs. It should not be used in lieu of full patient evaluation or solely relied upon to make or confirm a diagnosis.

It is designed to aid dental health professionals to review bitewing and periapical radiographs of permanent teeth in patients 12 years of age or older as a concurrent and second reader.

8. Summary of Substantial Equivalence:

The predicate device as well as the reference device and subject device are similar devices in the following ways:

- 1) *Intended use*: All devices are intended to be used to aid dental clinicians in their review of bone level measurements in bitewing and periapical radiographs of permanent teeth.
- 2) *Technology characteristics*: All devices employ computer vision and machine learning to output detections, use cloud-based environments to conduct processing, and display bone level measurements within a user interface with a graphical overlay over radiographs.
- 3) *Safety*: As both the candidate and predicate devices are software systems, neither pose a direct safety hazard to the patient. The primary hazards for both devices, subject and predicate, are potential inaccurate or misplaced bone lines/measurements which could result in a temporary non-serious injury. In the case of each device, users are not meant to rely solely on detection output for clinical decision-making.
- 4) *Clinical Performance*: All devices have undergone clinical studies which demonstrate statistically significant improvement in aided reader performance.

Table 1: Comparison of Second Opinion BLE with the predicate / reference devices.



K243230

	Subject Device Second Opinion® BLE	Primary Predicate Overjet Dental Assist K210187	Reference Device Videa Perio Assist K223296
Manufacturer	Pearl Inc.	Overjet, Inc.	VideaHealth, Inc.
Classification	892.2050	892.2050	892.2050
Product Code	QIH	LLZ	QIH
Class	Class II	Class II	Class II
Image Modality	Radiograph	Radiograph	Radiograph
Intended Use	Dental automated image processing software device to aid in bone level measurement dental radiograph review by HCP	Dental automated image processing software device to aid in bone level measurement dental radiograph review by HCP	Dental automated image processing software device to aid in bone level measurement dental radiograph review by HCP
Full IFU	Second Opinion® BLE is a radiological automated image processing software device intended to identify and display bone level measurements in bitewing and periapical radiographs. It should not be used in lieu of full patient evaluation or solely relied upon to make or confirm a diagnosis. It is designed to aid dental health professionals to review bitewing and periapical radiographs of permanent teeth in patients 12 years of age or older as a concurrent and second reader.	Overjet Dental Assist is a radiological semi-automated image processing software device intended to aid dental professionals in the measurements of mesial and distal bone levels associated with each tooth from bitewing and periapical radiographs. It should not be used in lieu of full patient evaluation or solely relied upon to make or confirm a diagnosis. The system is to be used by trained professionals including, but not limited to, dentists and dental hygienists.	Videa Perio Assist is a radiological semi-automated image processing software device intended to aid dental professionals in the measurements and visualization of mesial and distal bone levels associated with each tooth from bitewing and periapical radiographs. Measurements are made available as linear distances or relative percentages. It should not be used in-lieu of full patient evaluation or solely relied upon to make or confirm a diagnosis. The system is to be used by trained professionals including, but not limited to, dentists and dental hygienists.
Intended body part	Dental	Dental	Dental



Technology	Utilizes computer vision neural network algorithms, developed from open-source models using supervised machine learning techniques	Utilizes computer vision neural network algorithms, developed from open-source models using supervised machine learning techniques	Utilizes computer vision neural network algorithms, developed from open-source models using supervised machine learning techniques
Device Description	Detection & display of linear bone level measurements	Detection & display of linear bone level measurements	Detection & display of linear bone level measurements

9. Technological Comparison to Predicate and Reference Devices

The fundamental technological principle for both the candidate and predicate as well as reference device is the automatic detection and display of bone level measurements using machine learning.

The candidate and predicate / reference devices are technologically equivalent as follows:

- All are software devices designed to run on Windows operating systems.
- All devices are designed to process digital intraoral bitewing and periapical radiographs.
- All devices use neural network-based computer vision algorithms for bone level measurement display.
- All devices display measurements within the user interface with a graphical overlay on the radiograph.
- All devices produce near-instantaneous detection results.
- All devices are considered to be of “moderate” level of software concern.
- All devices passed all verification and validation testing requirements.

10. Assessment of Benefit-Risk, Safety and Effectiveness, and Substantial Equivalence to Predicate Device

Pearl demonstrated the benefits of the device through a standalone clinical study. The results of the study showed the system is both safe and effective for its intended use. When the probable benefits and probable risks of Second Opinion BLE are weighed against one another, the weight of benefits significantly exceeds that of risks. This judgment can be made based on review of the submitted materials showing that Second Opinion BLE meets the design verification and validation requirements. It is thus concluded that Second Opinion BLE can be considered safe and effective such that the device will aid users in the indicated user population in their radiographic review of bone heights.



11. Discussion of Non-Clinical Tests Performed

The device is a software-only device, so most testable characteristics common to other device types, including Biocompatibility/Materials, Shelf Life/Sterility, Electromagnetic Compatibility and Electrical Safety, Magnetic Resonance (MR) Compatibility, are not applicable to this device.

12. Discussion of Clinical Tests Performed

Clinical evaluation of Second Opinion BLE was performed to clinically validate the inclusion of a new bone level measurement tool in Second Opinion. The software modification is intended to assist dentists with their manual comparison of radiographs over time by providing automatic estimated bone level measurements in the bitewing and periapical radiographs. The effectiveness of Second Opinion BLE was gauged in a standalone performance assessment to validate the clinical usefulness of the added bone level measurement tool in Second Opinion BLE in intraoral bitewing and periapical radiographs of the permanent teeth of patients 12 years of age or older.

Clinical testing of the Second Opinion BLE software was performed utilizing retrospective clinical subject data from 229 bitewing and 167 periapical radiographs from 396 subjects. The demographics of the data set was as follows:

Subject Age	Count (%)
<30	63 (15.9)
30-<40	54 (13.6)
40-<50	79 (19.9)
50-<60	71 (17.9)
60-<70	73 (18.4)
70+	56 (14.1)

Gender	Count (%)
Female	160 (40.4)
Male	190 (48.0)
Other	44 (11.1)



Unknown	2 (0.5)
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No information about ethnicity was available from the dental radiographs

Second Opinion BLE utilized three US licensed dentists to label interproximal bone levels. The three readers provided measurements per CEJ-bonecrest, measurement differences that were clinically significant required an adjudication. These divergent measurements were then adjudicated by two U.S. Dental Radiologists.

Second Opinion BLE demonstrated an 87% (95% CI: 85%, 89%) precision value and 87% (95% CI: 85%, 89%) recall value for measurement of interproximal bone levels in periapical images and a 87% (95% CI: 86%, 88%) precision and 91% (95% CI: 90%, 92%) recall in bitewing images. Second Opinion BLE met the prespecified acceptance criteria of > 82% for both precision & recall for bitewing and periapical images.

Metric	Bitewing	Periapical
Precision	87%	87%
Recall	91%	87%

The mean interproximal bone level measurement of GT Second Opinion BLE and the mean absolute difference in millimeters (mm) by image type and overall are shown in the table below. The modeled mean absolute difference (95% CI) was 0.45 mm (0.37, 0.52) and 0.86 mm (0.83, 0.89) in periapical and bitewing images respectively. Second Opinion BLE met the pre-specified acceptance criteria of mean absolute difference < 1.5 mm in both bitewing and periapical images. In addition, the mean absolute difference remained statistically significant after the multiple comparisons adjustment as shown by all p-values < 0.05.

Based on the performance data and the application of general controls and special controls established for this device as published in the subject labeling, use of this device is unlikely to increase the rate of false negative or false positive diagnosis of dental pathologies as compared with the current clinical standard of practice. Further, possible misuse of the device does not present additional risks compared with the misuse of other types of radiological image processing devices.



13. Conclusions

Based on the information presented above, Second Opinion BLE and its primary predicate device, Overjet Dental Assist, are deemed to have similar intended uses as devices which aid in the detection & review of bone heights in dental radiographs. Second Opinion BLE's clinical trial results demonstrate that the device effectively performs as well as the primary predicate device.

Second Opinion BLE raises no new or different questions of safety or effectiveness, performs in accordance with its specifications, meets user needs, meets the intended use and therefore was found substantially equivalent to the predicate devices.